AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Generative Digital Bio Twin

Team ID: *GEN26-A0H-I161*

Team Name: *Byte ME*

Domain: *5*

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Presentation Outline

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Problem Statement

Developing climate-resilient crops currently relies on months of costly trial and error. Scientists need a faster, in-silico way to identify and evaluate genetic modifications that improve drought and heat resistance without reducing crop yield or quality, addressing the urgent global need to produce more food with limited resources.

Problem Understanding

- Agricultural Biotechnologists, AgTech R&D Companies, and Farmers seeking climate-resilient crop varieties.
- Unpredictable field stress, slow physical trial cycles (8–13 years), and unintegrated molecular vs. environmental data.
- High financial loss, late-stage failure rates (80–90%), elevated food security risk, and increasing resource scarcity.

Existing Gaps & Challenges

- CRISPR and bioinformatics tools focus on specific sequence edits but fail to predict whole-crop performance in the field.
- Multi-omics data and real-time agricultural IoT environmental data exist in disconnected silos.
- Modeling dynamic Gene $\times$ Environment ($G \times E$) interactions remains difficult with static conventional prediction models.

Our Proposed Solution

- Generative Bio Digital Twin: An AI-driven simulation platform integrating multi-omics data, gene editing, and climate data.
- Simulates plant response to stress factors (e.g., drought, high heat) and predicts crop yield and disease risk in real time.
- Enables researchers to test and compare candidate genetic edits computationally before conducting physical experiments.

Technical Approach

Multi-Omics & Field Sensors → Data Server → Deep Learning Model → Generative Bio Digital Twin Visualisation Dashboard

Biological and environmental inputs stream into deep learning engines to generate digital twin performance predictions.

Tech Stack

Frontend: React.js, Three.js

Backend: Python, PyTorch

Database: PostgreSQL / MongoDB

AI/ML, APIs, or Hardware: BioPython, CRISPOR, COBRApy, APSIM/DSSAT, IoT Sensors

Innovation & Uniqueness

- Combines molecular-level CRISPR analysis with macro-level climate and crop growth models in one unified platform.
- Generative digital twin offers an interactive 3D plant representation with real-time trait and stress visualization.
- Shifts crop R&D from slow physical trial-and-error to rapid, predictive *in-silico* screening.

Feasibility & Scalability

- **Feasibility:** Built on established open-source libraries (BioPython, COBRApy, APSIM) and standard GPU infrastructure.
- **Scalability:** Cloud HPC architecture allows expansion to multiple crop species, soil profiles, and regional climate scenarios.
- **Risks:** High complexity of $G \times E$ biological interactions; mitigated through continuous field trial data validation.

Cost for Product (Optional)

- Hardware & Infrastructure: GPU Workstation (32–64 GB RAM) and Cloud HPC compute nodes.
- Software & Sensors: Open-source modeling tools with local IoT sensor field deployments.
- Estimated Total: Low initial software prototype cost; eliminates millions in failed physical trial plot expenses.

Use Cases

- Use case 1: Drought Resilience Screening — Biotechnologists test candidate CRISPR edits in-silico under simulated prolonged drought.
- Use case 2: Yield vs. Stress Trade-off Analysis — R&D teams evaluate if drought-resistant genetic traits negatively impact net crop yield.
- Use case 3: Extreme Weather & Disease Alert — AgTech operators monitor real-time weather alerts and simulate plant immune system responses.

Impacts and Benefits

- Significantly shorter R&D cycles and reduced seed development costs for agricultural researchers.
- Enhanced food security and crop yield stability for farmers facing climate change and drought.
- Measurable reduction in water, land, and chemical resource waste through optimized crop breeding.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.4	Zero Hunger	Promotes resilient agricultural practices that increase productivity and crop yield.
SDG 13	Target 13.1	Climate Action	Strengthens adaptive capacity and resilience of crops to climate-related hazards.

Individual Member Contribution

Member	Contribution or responsibility
Jeffrey Immanuel J	Biology and research / Active research
Yazhini B	AI/ML and evaluation / Dataset preparation, baseline model
Jagan Sai Reddy	Crop Modelling / Real time scenario and feasibility
Gayathri T K	Research and review / Crop, drought traits and biological assumptions

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